The Answers

Note: the sample answers given here are very brief, touching on the salient points. The students are expected to elaborate around these points

1.

a)

i) What are switching surges in electric power transmission ?

[2]

Switching surges arise out of power circuit breaker openings/closings of lines. Usually breaking an inductive circuit manifests as large transient over voltage, closing of a capacitive circuit results in large transient charging currents. These last for several hundred milliseconds.

ii) How are they characterised ?

[2]

Normally several hundred kVs and hundreds of milliseconds

iii) How are they different from lightling surges ?

[2]

Lightning surges are in several kA and in micro seconds caused externally where as switching surges is caused internally.

iv) Why are they so important for network element design such power circuit breaker?

[2]

Insulation for breakers and all its live components are important and expensive, these transients impact on the requirement of the insulation volume and costs.

v) What kind of surge is most important in transformer protection design?

[2]

Mainly lightning , but occassioanly over voltages too.

b) This part relates to large synchronous machine construction

i) Why are the exciation windings housed on the rotor ?

[2]

It is low power circuit carrying currents and voltages at modest level (tens of amps and couple of 100 volts). The brush and slip ring arrangement can handle well.

ii) Why are the armature windings housed in the stator ?

[2]

Armature carries large currents (tens of thousands) at several kV. Stationary coil is convenient to handle rather than getting it from rotational system through brush slip ring arrangements.

iii) Why should the rotor be of solid construction ?

[2]

It has DC winding roatting with it , so no induced emf. Solid construction helps damping against oscillations.

iv) Why is the stator laminated ?

[2]

Power frequency AC supply currents flow through stator which interact magnetically with stator magnetic core giving rise to magnetic losses and so heating . Laminations reduces that losses.

v) Why are the stator conductors of hollow tubular structure?

[2]

Current flows more through surface of large conductor at power frequency. Hollow conductors help improve their cooling by allowing liquid coolant to flow through them.

vi) How is the frequency of stator voltage related to the rotor speed and the number of poles ?

[2]

*120*frequency / no of poles = Rotor speed (RPM)*

c) Usually the choice of generation voltage of a large synchronous generator is restricted between 15-25 kV. Narrate some reason to justify this choice.

[3]

Total cost of the generator depends on the cost of conductors and insulations – voltage lower than this means larger currents (thicker and coslier conductors) for a given MW capacity. Voltage higher than this means more insulation, larger size of the stator and so is the cost.

[25 marks]

2.

a)

i) Why do the steam turbine driven generators have round rotor?

[2]

High speed turbo machines are very efficient , so round rotor is good choice.

ii) Why do the hydro turbine driven generator have salient rotor ?

[2]

Usually they are low speed machines , so having salient poles helps to house large number of poles.

iii) Why is it not very useful to equip a hydrogenator with high gain fast acting speed control system without a transient gain reduction mechanism?

[4]

Transient gain reduction(TGR) helps to reduce the gain cross over frequency thus having increased phase margin . This helps stability of the system. In the absence of TGR the phase margin available at gain cross over frequency is very low because of RHP introduced by the turbine dyanmics.

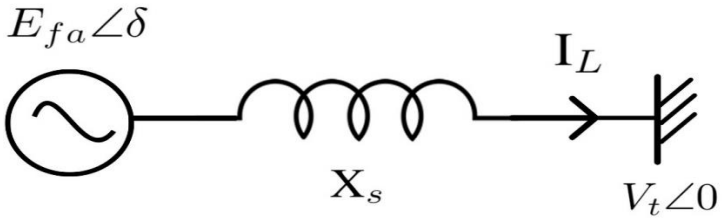
iv) How can a hydro generator favourably contribute to the rotor anagle stability ?

[2]

Saliency in the rotor produces reluctance torque which opposes the acceleration of the rotor during fault. This improves transient stability performance with larger critical clearing time for the breaker.

b)

(i) Starting from basic circuit equations, establish the power angle relation of a round rotor synchronous generator. You may take the help of the following circuit diagram of a generator connected to network with resistance neglected.



[7]

$$I = \frac{E_{fa}\angle\delta - V_t\angle 0}{jX_s} \quad [1]$$

$$S = V_t I^* = V_t\left[\frac{E_{fd}\angle -\delta - V_t\angle 0}{-jX_s}\right]$$

$$= \frac{V_t E_{fd}\angle 90-\delta}{X_s} - \frac{V_t^2\angle 90}{X_s} \quad [3]$$

$$\frac{E_{fa} V_t}{X_s}\sin\delta + j\left[\frac{E_{fa} V_t}{X_s}\cdot\cos\delta - \frac{V_t^2}{X_s}\right]$$

$$P = \frac{E_{fa} V_t \sin\delta}{X_s} \quad [3]$$

$$Q = \frac{E_{fa} V_t \cos\delta}{X_s} - \frac{V_t^2}{X_s} \quad [7]$$

c)

(i) The above generator winding is star configuration and is connected to the grid at 23 kV (line to line) voltage (*Vt*). The rated power is 500 MW, power factor is 0.85 (lagging). The synchronous reactance (Xs) is 1.8 pu on its rated base of 588 MVA. Find out at full output power:

ii) the excitation phase voltage in kV

[2]

iii) load angle (δ) in degrees

[2]

iv) load angle stability margin in degrees

[2]

v) total terminal reactive power in MVAR

[2]

[25 marks]

c) Rated. Base KV = 23
Base Pmm = 500 MW, base MVA = 588

2. Base impedance = $\frac{KV^2}{MVA} = \frac{23^2}{588} = .8997$

$X_s = 1.8$ pu, X_s in $\Omega = 1.8 \times .8997$

$I = \frac{MVA}{\sqrt{3} V_L} = 14760 A. = 1.6194 \Omega$

$E_{fd} = V_t + \sqrt{3} I X_s$

$= 56.97$ KV (L-L)

(i) E_{fd}/phase = 32.9 KV [2]

(ii) load angle = 38. Angle or E_{fn}. = 38.14 degrees [2]

(iii) Stability margin = 90 − 38.14 = 51.85 degrees [2]

(iv) total MVAr = 309 MVAr. [2]

3.

a) This part refers to synchronous machine:

i) What is a rotating reference frame in the context of synchronous machine modelling and analysis ?

[2]

It is a frame of reference of the observation that rotates at synchronous speed. This helps observing the voltage , current , flux and MMF in an easy way.

ii) What are the advantages of rotating reference frame over its static counterpart when analysing behaviour of stator rotor magnetic interactions ?

[3]

Inductance becomes time invariant and space invariant (irrespective of rotor position). The rotor and stator currents /volatges in steady state are all DC quantitities- very convenient to analyse.

iii) Why is one rotating reference frame not enough in the contexts of multimachine stability study ? [2]

Rotating frame is with respect to the rotor speed and angle position of that individual machine. In large network with multiple machines, each individual machine reference need to have a common reference – which is the network reference frame. This needs another transformation – usually known as Park to Kron transformation.

iv) What are stator transients in the contexts of synchronous machine operation?

[2]

During steady state the voltage induced on the stator winding is AC – expressed as a sinusoidal phasor. When fault appears and or machine speed deviates, transient currents /voltage appears in them which is different from steady power frequency quantities. These last depending on the point of inception of the fault, decaying dc components etc. In d-q domain they are treated as transformer voltage.

v) How are stator transients represented for stability studies in rotating reference frame?

[2]

Usually as time varying flux and current with appropriate relation through istator nductance and stator resistances.

vi) Why are stator/network transients traditionally neglected in first swing stability study for a large synchronous machine ?

[3]

They are very high frequency and also because of resistance they provide positive damping , so neglect of them always provides conservative answer as far as stability is concerned.

b) This part refers to asynchronous machine:

i) Why can not these stator transients be neglected in converter interfaced wind farm stability study?

[3]

Fequency of dyanmics leading to instability is lot higher (not electromechancial rather electromagnetic) , so ignoring them does not provide proper assessment of stability situation.

ii) Why is Type 3 (DFIG) cheaper than Type 4 (PMSM) for the same rated output capacity ?

[3]

Type 4 has full conveter – which is lot costlier than that used in Type 3 which only handles less than 30% of the total capacity

iii) Why is angle stability not as concerning as oscillatory or synchronisation stability in the operational context of grid connected wind farm?

[3]

There is no electromechanical load angle present in the context of wind farm. Here synchronisation issues are far more relevant and so is the pure electrical oscillations.

iv) How is the wind *wake* represented for fast frequency response provision

[2]

Usually a fraction or simple formula used to adjust the net output from Wind turbine in the subsequent rows while moving along the direction of the wind

[25 marks]

4.

a)

i) Starting with the definitions, narrate how the capacity factor in a wind turbine differ from the power conversion coefficient (Cp) ?

[4]

Capacity factor is the actual annual energy produced as fraction of maximum energy that could have been produced at rated output all through out the year. The power conversation co-efficient is the fraction from the kinetic energy of the wind getting converted to mechanical energy in the turbine. The former is dependant on the weather and wind velocity profile , the later is on the turbine design and control.

ii) Why do the offshore wind turbines of similar capacity produce more energy over a year over their onshore counterpart?

[2]

The average wind velocity offshore always remains higher and uniform than that for the onshore situation. The wind profile also fluctuates often on the land than that of in the sea.

iii) What is the tip speed ratio in the context of wind energy conversion and control ?

[2]

It is the ratio between the linear velocity of the blade at the tip to the wind velocity. This ratio can be adjusted by adjusting turbine speed

iv) Why is it so important to have the optimal tip-speed ratio in wind turbine operation ?

[2]

Below rated wind speed , the power conversion coefficient (Cp) is influenced by this ratio. At a certain ratio (known a soptimal tip speed ratio), the maximum power is captured. This is known as maximum power point tracking (MPPT)

v) When is the tip-speed ratio control not effective ?

[2]

At a rated or higher wind velocity it is not effective as the control switches over to pitch/stall control.

b)

i) What are the fundamental differences between the pitch and stall control in wind farm operation ?

[4]

Pitch control adjust the blade face or angle of attack such that part of wind is not faced by the blade, thus reducing the amount of power captured. The stall control is to face the

full wind when it is blowing at higher than rated , thus creating turbulence on the leading surface with drop in lift and increase in drag. This also reduces net power captured.

ii) How are the pitch control of WTGs modelled for stability studies ? You may take help of suitable block diagram to answer this question

[3]

It is modelled as a slow first order block with ramp and anti windup limit as well as with a dead band. The answer is expected to have some block diagram.

iii) How are the stall control mechanism incorporated in modern wind turbine fleets.

[2]

Active stall produces lots of stress on the wind blade, bearing and hub. This is discontinued. Only passive stall control is in place for modern turbines where the curvature of the blade aero foil structure is arrived at by design. This is influenced by many factors such as wind speed, direction , blade length etc.

iv) How is the rotor flux of a permananent magnet synchronous machine (PMSM) type wind turbine modelled for stability studies ?

[2]

There is no current as it is PMSM. Usually a sinusoidal spatial distribution with maximum value of flux e.g. $\varphi = \varphi_m sin\theta$

v) How are the turbine and generator shafts of a DFIG (Type 3) modelled for stability studies?

[2]

Usually by first or second order differential equations with speed , twist angle and mechancial torque as done in equation of rotational motion. Mostly gearbox is represented through a scalar coefficient.

[25 marks]